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Pier 55, NYC: A Case Study for the Future of Design, Documentation and Fabrication

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Abstract

The Pier 55 project in New York City represents an achievement in design, documentation, fabrication and construction achievable only through recent advances in construction technology. Pier 55 is a new park built over the Hudson River constructed from complex precast concrete. It is a one of its kind pier with a signature design by the Heatherwick Studio that undulates in elevation and is structurally composed of tulip shaped concrete "pots". Heatherwick's vision required significant collaborative efforts by all involved to define a geometry that satisfied the often-competing needs for prefabrication efficiency, durability, accessibility, design aesthetics and construction feasibility. Arup and Heatherwick developed parametric tools to automate much of the design process so that multiple iterations of geometry could be tested and refined to find optimal solutions. Initial scripts to define surface geometry of the "pot" structures for coordination evolved into additional scripts which created analysis models, full structural geometry, and shop drawing level documentation. As the project moved into construction, Arup and the fabrication team at Fort Miller precast concrete manufacturer and Fab3 steel fabricator utilized the models and scripts generated during the design process for direct digital input of the structural geometry to create complex CNC-milled foam formwork, 3-dimensional rebar documentation, and documentation and digital fabrication of steel components required for assembly and erection of the various pieces by Weeks Marine. This paper will discuss significant innovations including using sophisticated parametric modeling to digitally design, document, fabricate and construct geometrically complex structures.

Keywords: pier, precast, parametric modeling, script, automation, digital fabrication

1 Introduction – A New Park on Hudson River

1.1 Project vision

Pier 55 is an artificial island that is being built on the Hudson River west side of Manhattan to provide a unique park experience for the New York public. It will be owned and maintained by the public agency Hudson River Park Trust, and has been funded primarily by Barry Diller as the private donor and cultural program operator.

Over the years, Hudson River Park Trust has successfully renovated old piers into public parks, recreational facilities or office and retail complexes. Pier 55 will revitalize this section of Hudson River Park with almost three acres of new public park space featuring lush greenery and a diverse array of world-class arts, educational and community programming.[1]

Under the overall design management by the prime consultant Mathews Nielsen Landscape Architect and in collaboration with the U.K.-based designer Heatherwick Studio, Arup was selected to realize the project vision for all engineering and consulting disciplines. Standard Architects is the executive architect and Mueser Rutledge Consulting Engineers is the marine engineer responsible for pile design and the Esplanade structure. Hunter Roberts Construction Group is the construction manager with Gardiner & Theobald acting as the owner's representative.

Figure 1. Architect's rendering of Pier 55[2] ©Pier55

1.2 Challenges

The signature design of Pier 55 which sets it apart from other projects with complicated and curved geometry, also imposes the most significant challenges for structure and construction. The geometry was not possible or practical to convey via conventional 2D documentation, so the team adopted an open approach to 3D modelling where we quickly realized that the only way to ensure geometry of construction would align on site would be a common design and construction model with the use of direct to manufacturing digital fabrication techniques for all the complex geometry elements.

2 Design Evolution – Parametric Design and Automation under True Collaboration between Architects and Engineers

2.1 Architectural vision realized with considerations for constructability, accessibility, and durability

Construction on the Hudson faces challenges due to the marine environment which typically push designs towards prefabrication to remove the need of the falsework. Pier construction generally consists of piles placed in the river bed, topped with precast piles caps and planks, and tied together with a cast-in-place concrete slab. These constraints made precast the right choice for this project, but precast technology would need to push past current convention to meet the architectural vision. A unique structural system has been proposed which is structurally effective, constructible, and, at the same time, visually intriguing.

Avoidance of a rectilinear column grid was one of the key aesthetic desires. To balance this desire with the cost constraints requiring that precast

concrete be of repetitive shapes, a unique tiling geometry called the 'Cairo Pentagon' pattern was adopted, which consists of identical but rotated and nested pentagons with two 90° corners and three 120° corners. With this, a seemingly irregular pile layout is achieved while the regularity still exists for the elements.

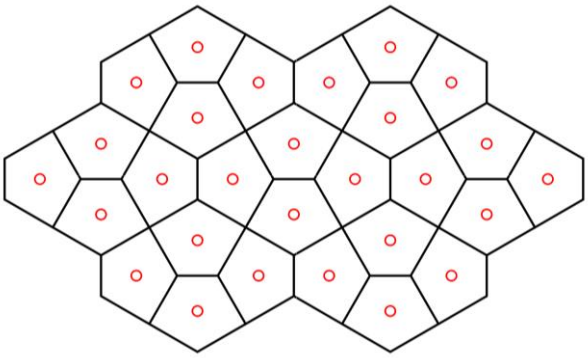

Figure 2. Cairo Pentagon tiling with pile locations

Instead of a thin flat deck structure, 'pots' made with architecturally exposed precast concrete are erected on top of each driven pile. These pots are connected by a topping slab and create a single monolithic deck structure that receives the large volume of soil required for the vibrant landscape.

To create the topography for landscaped hills, strolling paths and overlooks, the deck and supporting pot structure itself varies in elevation following the top surface contour rather than piling up soils and foams from a flat deck. Finish surface elevation of the pier varies from 15' to 65' above the sea level.

2.2 Structural system

2.2.1 Piles

Bedrock elevation varies significantly across the pier with elevations as high as 70' below water level near shore and as deep as 240' towards the river centre. 36" Diameter precast prestressed piles were selected in order to carry heavy vertical loads from the pot and deck structure and landscape. Among 281 total piles, the maximum length is 260 ft., and the maximum axial capacity is designed as 325 tons. The length of the precast portion is limited to 128', with a steel HP14 extension as required to reach bearing strata on rock below. Two accessways from the Esplanade to the Pier utilize 46 of 24" square prestressed and precast concrete piles. Pile design is by Mueser Rutledge Consulting Engineers.

2.2.2 The Pot

Majority of the deck structure consists of 132 unique 'pots'. The typical 'pot' is 15' tall and approximately 400 ft^2 on plan area. Each pot is composed of 4~6 'petals', with thickened side struts and bottom edge beams, that sit on a central 'column head'. The column head seamlessly transitions from the 'pot' bulbous curved shape to the cylindrical pile geometry. The center of the column head acts as a vertical support for planks and 'star beams' above, which span radially from the center to the pot corners. Petal interiors typically transition from sloped to vertical moving away from the column head. Stainless steel embedded plates and welded external plates connect the petals to each other and the column head.

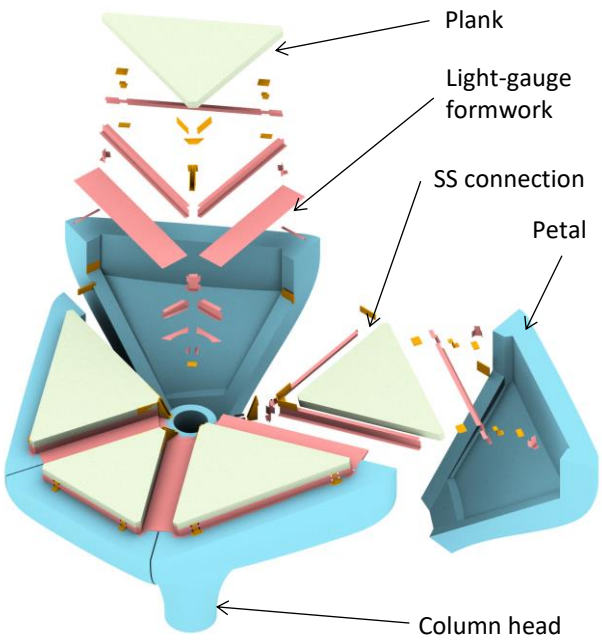

Figure 3. Exploded view of a Pot

The hollow piles are connected to the 'pots' through a cast-in top plug and guide column. The steel guide column is cast into the pile via a ~7' long plug of cast in place concrete and then used to both orient the pot in plan and provide temporary stability for the pot. Once the pot is erected and properly shimmed, the hollow core is filled with concrete to achieve an integral connection between the pile and pot structure.

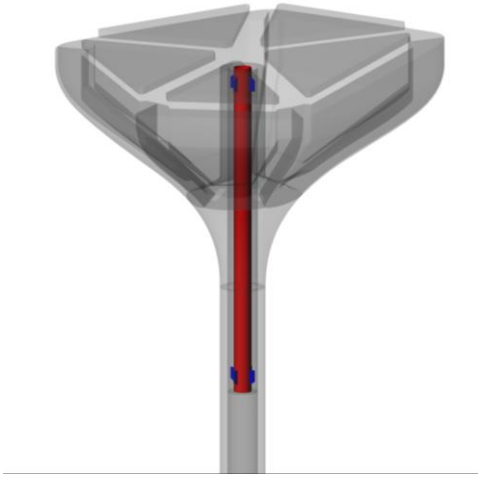

Figure 4. Guide column in the pile

Triangular planks span between the column extension and the upper edge of pot petals to act as stay-in-place permanent formwork for an in-situ topping slab that ties the pots together and holds the landscape above. The edges of the planks support light gauge bent sheet metal which acts as stay-in-place formwork for the cast-in-place star beams which carries the plank and slab loads back to the pot edges and central column extension. The star beams also act as part of a moment frame for overall lateral stability of the pier.

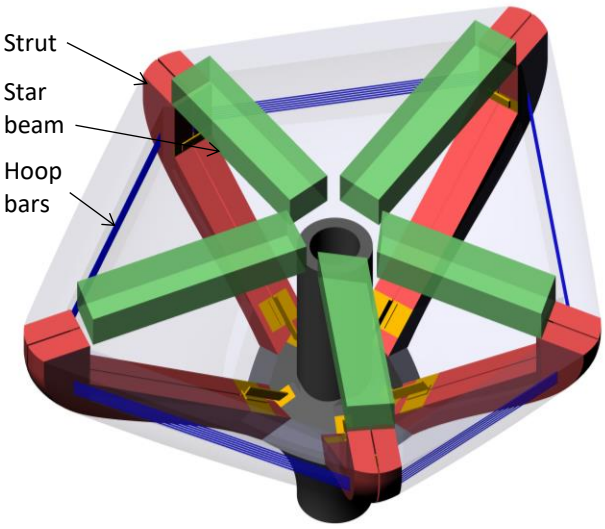

Figure 5. Structural composition of a Pot

2.2.3 The Flat Pier

The center portion of the pier is predominately framed with more conventional rectilinear precast concrete bents and planks with cast-in-place topping in order to minimize cost and leave room for critical program areas such as back of house, undercroft, public restrooms, utility rooms, and actors changing rooms. This flat pier area also supports the 750-seat amphitheater with raker walls and stepped seating.

2.2.4 Overall system

The resulting structure is a highly indeterminate moment-frame structure, and for the seismic design, detail requirements of the intermediate moment frame are applied. Much of the pot and petal structure is actually controlled by construction stresses during erection, so the design team needed to consider construction staging in the design of these elements. The permanent design considers vertical loads due to self-weight, superimposed landscape and finish loads, pier live loads, and theater live loads, as well as lateral loads from temperature, wind, seismic, ice, and wave loads.

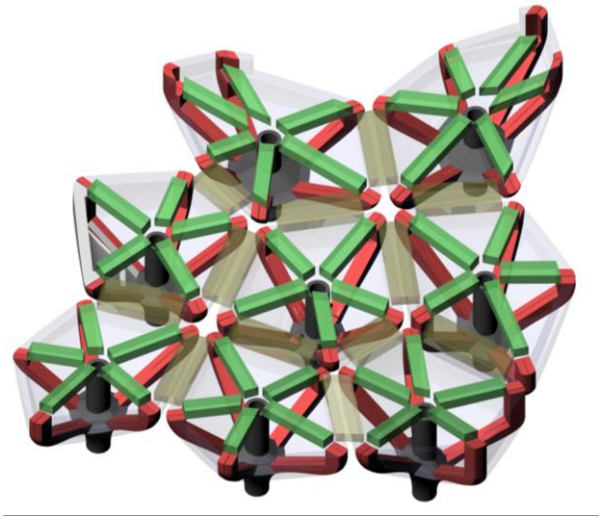

Figure 6. Inter-connected pots

As the pier will be maintained by the State agency, the durability of the structure is a major consideration, and a number of measures have been applied including corrosion inhibitor admixture in the concrete, entrained air, rapid chloride ion penetration test, freeze-thaw test, stainless steel connections, epoxy coated rebars, and generous clear cover for rebars.

2.3 Geometry generation of the Pot

The outside surface geometry of the pot is generated by the architect through a parametric script. Base rules of geometry are defined through discussions between architect and engineer to balance complexity with architectural vision. For example, the petal outer surface generally has a one-way curved surface except edges where it rolls up to make the petal joint smooth, so the horizontal rebars can be straight with splicing edge bars that are hooked and anchored in the cages of the strut bars. Internal surfaces are generated by Arup using another set of parametric algorithms to meet the required structural criteria and are geometrically simple to facilitate the conventional formwork fabrication for the interior surfaces. The joint between the column head and the petals is also defined by Arup making sure the connection using steel plates is feasible and the drainage of incidental water inside pot is considered.

2.4 Rationalization

Although the overall layout and the plan shape of pots follow the Cairo-Pentagon pattern, fitting them into a square overall plan with relatively straight edges and providing varying elevations across the top surface resulted in many different pot shapes. 39 different pot types resulted in 39 column head forms and 192 unique curved petal forms used to generate 132 pots and 656 petals. The curved exterior faces at the bottom portion of the pot are typical within pot types and the elevation variation required to meet the topography of the pier are achieved through pouring varying height 'extension walls.' So, while each pot and each petal are unique, repetition of formwork was possible to a great extent. A few very special and complex pot types particularly in areas over the southern accessway and near the amphitheater are only used once.

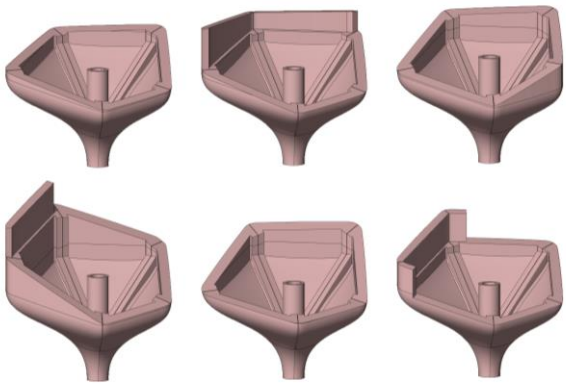

Figure 7. All pots in Type 8

2.5 Parametric generation of analysis, design and fabrication models

2.5.1 Analysis model

Arup used an extension to the same parametric geometry generation script to generate highly detailed analysis model needed to design the rebar for all precast elements as well as connection plates and embeds. All design elements are explicitly modeled and custom final member design post processing tools were created to calculate the required rebars, steel plate sizes, and the number of studs from model force outputs directly and quickly generate the results.

2.5.2 Connections and light gauge formwork model

Pot precast elements feature 7 distinct connection types with multiple instances of each type based upon petal and plank geometry meaning an average of 35 pieces that may have unique set-out, unique geometry, and unique plate size and weld prep definitions dependent upon overall pot geometry and loading. Over 15,000 plates are defined and set-out parametrically.

The light gauge sheet metal formwork features 12 distinct elements to form the star beam elements. Slab geometry, which drives plank and star beam geometry, is essentially unique across all pots resulting in over 7,700 elements in total that are defined and set-out parametrically. The light gauge includes bolt hole locations and cutouts to avoid conflict with the steel connections.

2.5.3 Rebar model

Also generated in 3D using parametric scripts by Arup are the rebar models for petals and column heads. For the mockup construction, the contractor manually modeled the rebars in those curved face elements and it took significant resources. This manual process was untenable for the entire project, so Arup was hired to directly model all rebars using scripts, and the final model provided to the contractor has all rebars modeled with the correct size, locations, splices, bending radius and hooks.

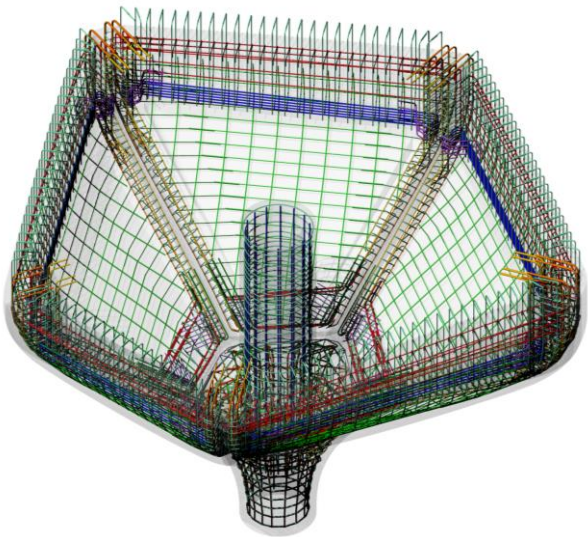

Figure 8. 3D rebar model

2.5.4 Coordination

All design team members worked in 3D using either Rhino or other BIM software, and coordinated trades in 3D as well. A Navisworks model was created by importing each team member's 3D components for coordination, clash detection and for contractor's information and reference.

2.6 Design deliverables in 3D

All surfaces of precast elements, exact shape and dimensions of stainless steel components and bent shape of light gauge stay-in-place formwork steel, and rebars in the complex petals and column heads are completely generated by the scripts with little manual adjustment, and the resulting models in the 'as-in-place' format in which all components are in their final locations in 3D and the 'arrayed' format in which all components are laid out in 2D surface are provided to the precast and steel fabrication contractors with full geometry for use in shop drawing production and final fabrication.

3 Fabrication – Rhino to Reality

3.1 Milled foam formwork

The doubly curved surfaces represented a significant challenge that conventional formwork (plywood/steel) could not achieve due to cost, schedule, and tolerances. The design and construction team concluded that digital fabrication using milled foam with surfaces generated from previous models was the only feasible approach.

A rigid foam block is milled by CNC machine for the outer petal surface. It is then sprayed with a polyurethane coating which hardens and allows for the form to be used multiple times. The process, foam and coating materials, stripping agents, and mix design was tested by Fort Miller over a number of mockups and prototype petals, including two full pots. Final pots erected on site indicate that the milled foam formwork can provide a free form (virtually any geometry) face of concrete with specific surface texture using coatings in a large scale with joints that align to an extent that would have been impossible with conventional formwork fabrication techniques.

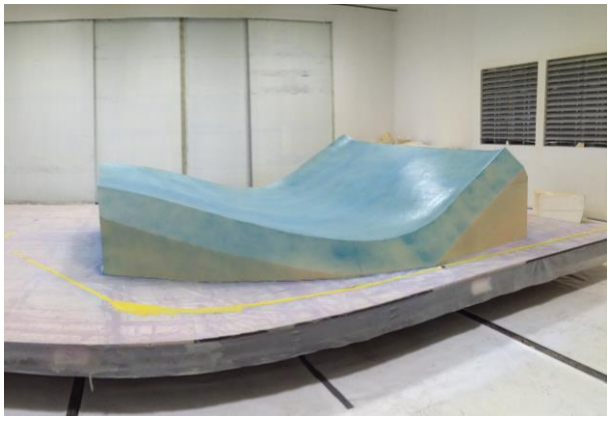

Figure 9. Milled foam with primer coating

3.2 Rebar bending information from 3D model

As mentioned in 2.5.3 Arup took on reinforcement modelling which kept the project feasible. As the design team was already providing digital models for shop drawings and foam milling, we further blurred the boundaries between typical designer and contractor scopes and utilized parametric scripts to efficiently generate full 3D rebar shop drawing models for both fabrication and placement. Each rebar piece is bent per shop drawing and is tagged with unique identifier for the correct placement.

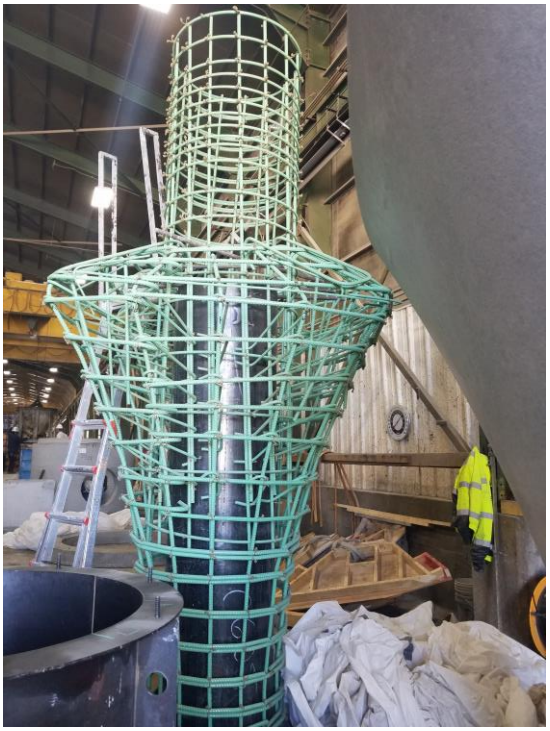

Figure 10. Fabricated bars with tags (above) and assembled (below) for a column head

3.3 Submittals in 3D

Any modifications that the contractor needed to make for means and methods were memorialized in shop drawings. In many cases, the shop drawings were reviewed in 3D.

4 Conclusion

Pier 55 is groundbreaking departure from conventional pier construction and utilizes the latest advances in digital design and fabrication technology to successfully deliver a structure that is both functional and playfully sculptural at the same time. The major challenge of generating, engineering, communicating and fabricating of the complex geometry was overcome with intensive use of parametric scripts, automation, digital data communication in the design phase, and the use of milled foam formwork, bending rebar using a detailed 3D model in the fabrication phase. The collaboration between all parties to deliver this project was critical in achieving a successful outcome for future generations to enjoy. When complete, Pier 55 will be a new icon of the west side, and visible reminder of what can be achieved when the conventional scope boundaries of contractors and designers are blurred and project participants come ready to collaborate.

Figure 11. Construction site as of April 2019

5 Acknowledgements

Arup is thankful for the opportunity to participate in the design of this unique and challenging project in New York City, and the collaboration with exceptionally talented clients, designers and contractors

6 References

[1],[2] www.pier55.com